

```
1 import java.util.Scanner;
2
3 public class Main {
4
5     static boolean isSafe(int[][] arr, int x, int y, int n) {
6
7         // Check column
8         for (int row = 0; row < x; row++) {
9             if (arr[row][y] == 1) {
10                 return false;
11             }
12         }
13
14         // Check left diagonal
15         int row = x;
16         int col = y;
17
18         while (row >= 0 && col >= 0) {
19             if (arr[row][col] == 1) {
20                 return false;
21             }
22             row--;
23             col--;
24         }
25
26         // Check right diagonal
27         row = x;
28         col = y;
29
```

```
30 while (row >= 0 && col < n) {  
31     if (arr[row][col] == 1) {  
32         return false;  
33     }  
34     row--;  
35     col++;  
36 }
```

```
37  
38 return true;  
39 }
```

```
40  
41 static void printBoard(int[][] arr, int n) {  
42
```

```
43     for (int i = 0; i < n; i++) {  
44         for (int j = 0; j < n; j++) {  
45  
46             if (arr[i][j] == 1)  
47                 System.out.print("[Q]");  
48             else  
49                 System.out.print("[ ]");  
50         }  
51         System.out.println();  
52     }  
53
```

```
54     System.out.println();  
55 }
```

```
56  
57 static void nQueen(int[][] arr, int x, int n) {  
58
```

```
59     // All queens are placed  
60     if (x == n) {  
61         printBoard(arr, n);  
62         return;  
63     }  
64
```

```

65 // Try every column
66 for (int col = 0; col < n; col++) {
67
68     if (isSafe(arr, x, col, n)) {
69
70         // Place queen
71         arr[x][col] = 1;
72
73         // Recursively place remaining queens
74         nQueen(arr, x + 1, n);
75
76         // Backtrack
77         arr[x][col] = 0;
78     }
79 }
80
81
82 public static void main(String[] args) {
83
84     Scanner sc = new Scanner(System.in);
85     System.out.println("Enter the number of queens:");
86     int n = sc.nextInt();
87
88     int[][] arr = new int[n][n];
89
90     nQueen(arr, 0, n);
91
92     System.out.println("-----All possible solutions-----");
93
94     sc.close();
95 }
96 }

```

Enter the number of queens:

4

[] [Q] [] []

[] [] [] [Q]

[Q] [] [] []

[] [] [Q] []

[] [] [Q] []

[Q] [] [] []

[] [] [] [Q]

[] [Q] [] []

-----All possible solutions-----

...Program finished with exit code 0

Press ENTER to exit console.